

APIs 101: A Beginner's Crash Course



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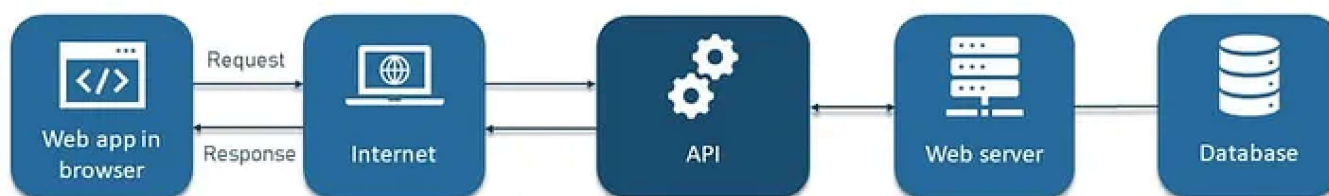


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What is an API?

An API, or Application Programming Interface, is a set of rules and protocols that enables software applications to communicate with each other. It facilitates the exchange of data, features, and functionalities between different applications, streamlining and expediting software development by allowing developers to integrate existing data and services from other applications instead of building them from scratch.

HOW API WORKS



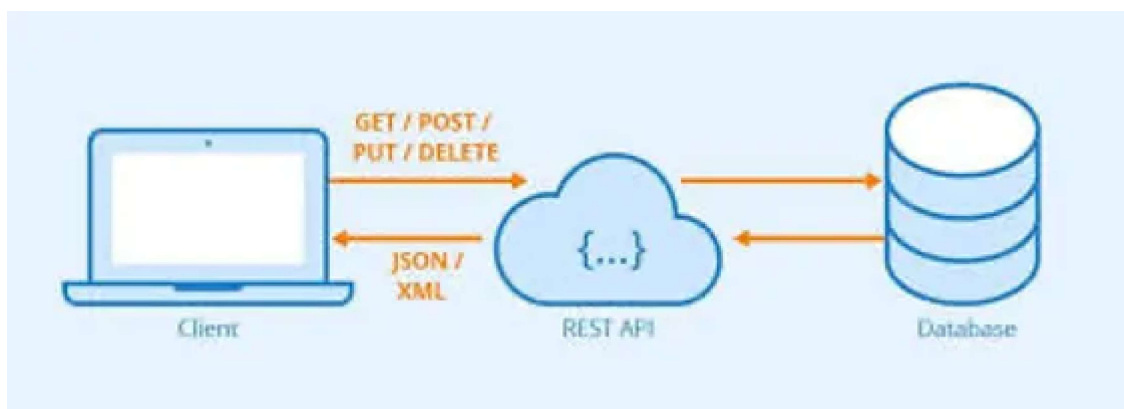
For example, Zomato, a popular food delivery service, uses APIs to track order status in real-time. When you place an order, Zomato communicates with the delivery partner's system via APIs to fetch updates on delivery status, location, and estimated arrival time. This information is presented in the Zomato app without directly accessing Google Maps; instead, it relies on APIs from delivery partners or location services. This approach allows Zomato to provide relevant updates while keeping the internal workings of the delivery partner's system hidden.

Types of APIs

APIs can be categorized based on their architecture and usage. The main types include REST, SOAP, and GraphQL.

REST APIs

REST (Representational State Transfer) APIs use the HTTP protocol to communicate between the server and the client. They rely on standard HTTP methods such as GET, POST, PUT, and DELETE to perform operations on resources, which are represented in JSON format.



Key HTTP Methods in REST APIs:

1. **POST**: Adds data to the server. For example, creating a new user account.
2. **GET**: Retrieves data from the server. For instance, fetching user details.
3. **PUT**: Updates existing data on the server. An example is updating user profile information.
4. **DELETE**: Removes data from the server. For example, deleting a user account.

For example

- **POST** request: To create a new Google map location
- **GET** request: To fetch created Google map location data
- **PUT** request: To update/edit created Google map location data
- **GET** request: To check whether Google map location is updated or not
- **DELETE** request: To delete Google map location

SOAP APIs

SOAP (Simple Object Access Protocol) APIs use XML messages to facilitate communication between applications over a network. They follow strict standards and rules, making them highly secure. SOAP APIs are often used in scenarios where security and transaction compliance are critical, such as payment processing and customer management.



GraphQL APIs

GraphQL APIs allow users to request specific data from the backend service in a single API call. This flexibility reduces the number of requests needed and improves efficiency by allowing clients to specify exactly what data they need



Benefits of Using an API

- **Flexible Information Transfer:** APIs facilitate seamless data exchange between applications.
- **Customization:** APIs can adapt protocols, functions, and commands as needed.

- **Efficiency:** Automate content publishing across multiple channels, improving operational efficiency.
- **Layered Information Distribution:** APIs enable the creation of app layers to distribute information to different audiences.

What is Flask Framework?

Flask is a Python web micro-framework used to create APIs with ease. It allows developers to define URLs (routes) that trigger specific functions in the code. These functions handle incoming requests, perform actions (such as calculations), and return responses in JSON format.

Example: Addition of Two Numbers Using Flask

Here's a simple example of creating an API endpoint that adds two numbers using Flask:



(Flask API for adding two numbers)



(Postman call for the above API)

Breakdown of the Code

- **Imports:** Necessary tools from Flask to handle web requests and convert data to JSON.
- **Flask App:** Creates a Flask application instance.
- **API Route:** Defines a route (`/add``) that listens for POST requests.
- **add_numbers Function:** Handles the POST request, retrieves data, checks for validity, calculates the sum, and returns the result in JSON format.

API Encryption

API encryption is a critical security measure that encodes the data exchanged between a client and an API to ensure confidentiality and integrity. The primary goal is to prevent unauthorized access, interception, and tampering.

API Encryption Process



1. **Client Request:** The client application prepares a request to send to the API.

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2. **TLS Handshake:** Establishes a secure connection using HTTPS, involving the exchange of encryption keys.

3. **Data Encryption:** The client encrypts the request data using the encryption key.

4. **Transmission:** Encrypted data is transmitted over the secure TLS connection.
5. **Server Decryption:** The API server decrypts the received data.
6. **Processing:** The server processes the request and prepares a response.
7. **Encrypt Response:** The server encrypts the response data.
8. **Encrypted Response:** The encrypted response data is transmitted back to the client.
9. **Client Decryption:** The client decrypts the received response data.
10. **Result:** The client processes the decrypted response and displays the information to the user.

TLS Handshake

The TLS (Transport Layer Security) handshake establishes a secure connection between a client and a server, ensuring data is encrypted and secure. It involves verifying identities and exchanging encryption keys.

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decryption. It's efficient but requires secure key management.

- **Asymmetric Encryption:** Uses a pair of public and private keys. Data encrypted with the public key can only be decrypted with the private key.

This is more secure but slower.

AES (Advanced Encryption Standard)

AES is a widely used symmetric encryption algorithm known for its speed and security. It employs secret keys of varying lengths (e.g., 128-bit, 256-bit) to encrypt and decrypt data.

JWT (JSON Web Token)

JWT (JSON Web Token) is a mechanism for verifying the authenticity of JSON data to ensure its integrity during transmission or storage. Each JWT is signed using cryptography to confirm that its contents have not been altered.

A JWT consists of three main components:

1. **Header:** A JSON object that typically contains:

- Type of token (JWT)
- Encryption algorithm used (e.g., HMAC SHA256, RSA)

2. **Payload** (also known as claims): A JSON object containing transmitted data, such as:

- User information (e.g., username, email address)
- Session data (e.g., IP address, time of last login)
- Authorization permissions (e.g., roles or groups the user belongs to)

3. Signature:

- Created by signing the Base64Url encoded header and payload
- Uses a secret key and a specified algorithm
- Verifies the sender of the JWT
- Ensures the token's integrity

API Authentication vs. Encryption



Best Practices for API Development and Testing

Using Postman for API Testing

Postman is a tool that simplifies the process of testing APIs by allowing developers to send requests and examine responses.

API requests: An application or server can receive an API request, which is a message asking a server to provide data or a service. It consists of the following elements.

1. **Endpoint:** The place that APIs send requests and where the resource lives, is called an endpoint. It is a dedicated URL that provides access to a specific resource.
2. **Method:** The particular actions the client wishes to take on the URL resource are known as the request methods. Standard HTTP methods, which can be used to get, create, update, and delete data, can be used to access REST APIs.
3. **Parameters** — Parameters *are key-value pairs* passed to an API endpoint (URL) to provide instructions for the API server to process. Parameters can *specify variable elements of a URL path, query parameters, headers, or a request body*.
4. **Request headers** — request headers used to provide more information for a server to process an API request.

Key Postman Practices

1. **API Automation:** Automate API testing to ensure consistency and coverage.
2. **Write Assertive Tests:** Use assertions in test scripts to check response status codes, data structure, and expected values.
3. **Reuse Code Logic:** Encapsulate logic into reusable functions to improve maintainability.
4. **Manage Test Data Effectively:** Plan how to handle test data for different scenarios, ensuring proper data isolation and reset mechanisms.

Conclusion

APIs play a crucial role in modern software development by enabling seamless communication between applications. Understanding how to create and secure APIs, as well as following best practices for testing, is essential for developers. Whether using REST, SOAP, or GraphQL, ensuring efficient and secure data exchange is paramount to delivering reliable and robust applications.

API

Api Gateway

Api Development

**Written by MUJ ACM**

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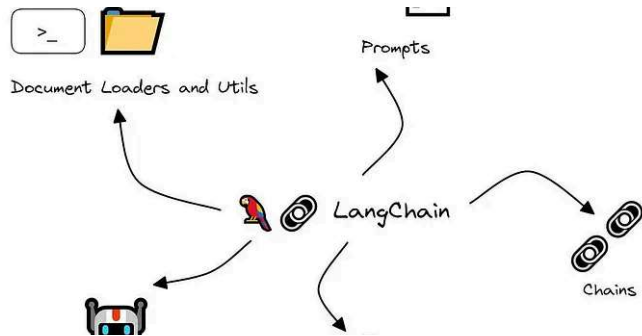
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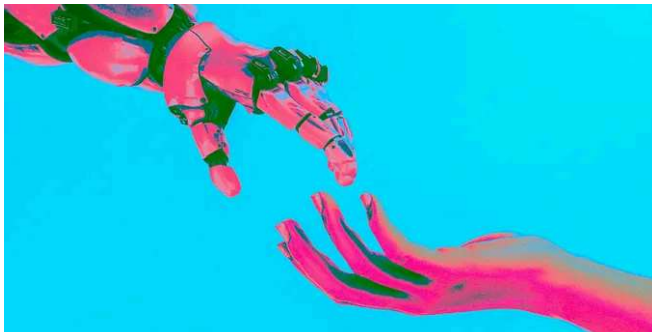


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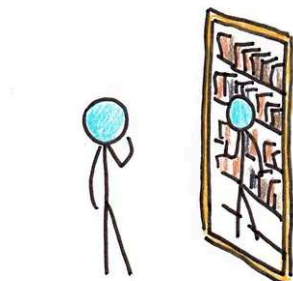
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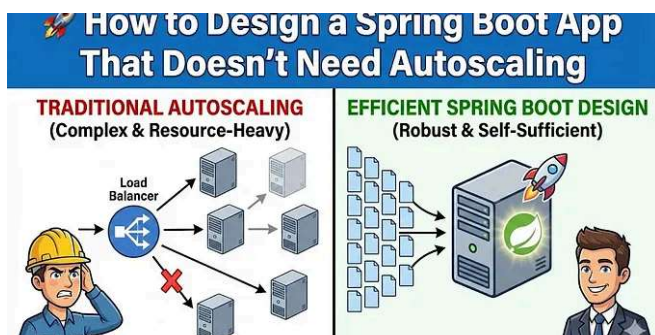


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